SMART WASTE MANAGEMENT USING IOT

A Project Report Submitted in Partial Fulfillment of the Requirement for the Degree of

BACHELOR OF TECHNOLOGY

IN

CIVIL ENGINEERING

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CERTIFICATE

This is to certify that the Project Report entitled "SMART WASTE MANAGEMENT USING IOT" that is being submitted by

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In partial fulfillment of the requirement for the award of B.Tech in Civil Engineering in the **RAJEEV GANDHI MEMORIAL COLLEGE OF ENGINEERING AND TECHNOLOGY,**Nandyal (Affiliated to J.N.T University, Anantapur) is a bonafide record of confide work carried out by him under our guidance and supervision. The results embodied in this technical report have not been submitted to any other university or institute for the award of any Degree.

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Dedicated to my beloved parents, and teachers who have worked hard throughout my education.

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Abstract

In many places, it can be seen that the Municipal garbage bins are overflowing and they are not cleaned at proper time. As a result of which the consequences are severe. It includes overflow of garbage which results in land pollution, spread of diseases. It also creates unhygienic conditions for people, and ugliness to that place. There should be a system that can monitor the bin and can give the information of filling of the bin to the municipality using wireless sensor network so that the bin can be cleaned on time and the environment can be safeguarded. The Smart waste management system that identifies fullness of the bin using a wireless sensor network (WSN). The system provides a web interface to the cleaning authority so that they can monitor and clean the garbage bin. In this ultra-sonic sensors are used to determine the status of the dust bins. The status is read by Node-MCU and transmitted to the cloud server (THING SPEAK). The server controlled in the municipality office will read this data and determine which dust bin is to be emptied or full and timely inform to driver to collect waste.

Keywords:Municipal garbage bin,Ultrasonic sensor,NodeMCU,ThingSPEAK,Monitoring.

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Chapter 1

Introduction

1.1 Introduction

Waste management are the activities and actions required to manage waste from its inception to its final disposal. This includes the collection, transport, treatment and disposal of waste, together with monitoring and regulation of the waste management process. One of the main concerns with our environment has been solid waste management which impacts the health and environment of our society. A large portion of waste management practices deal with municipal solid waste (MSW) which is the bulk of the waste that is created by household, industrial, and commercial activity.
There are different types of waste:

1. **Liquid waste.**
2. **Soild Rubbish.**
3. **Organic waste.**
4. **Recyclable waste.**
5. **Hazardous waste.**

1.Liquid waste.

- Liquid waste is commonly found both in households as well as in industries. This waste includes dirty water, organic liquids, wash water, waste detergents and even rainwater.

- You should also know that liquid waste can be classified into point and non-point source waste. All manufactured liquid waste is classified as point source waste. On the other hand, natural liquid waste is classified as non-point source waste.

- Tt is best get in touch with waste removal experts, such as 4 Waste Removals, to dispose of liquid waste properly.

2.Soild Rubbish.

- Solid rubbish can include a variety of items found in your household along with commercial and industrial locations.
- Solid rubbish is commonly broken down into the following types:

- **Plastic waste -** This consists of bags, containers, jars, bottles and many other products that can be found in your household. Plastic is not biodegradable, but many types of plastic can be recycled. Plastic should not be mix in with your regular waste, it should be sorted and placed in your recycling bin.
- **Paper/card waste -**This includes packaging materials, newspapers, cardboards and other products. Paper can easily be recycled and reused so make sure to place them in your recycling bin or take them to your closest Brisbane recycling depot.
- **Tins and metals -** This can be found in various forms throughout your home. Most metals can be recycled. Consider taking these items to a scrap yard or your closest Brisbane recycling depot to dispose of this waste type properly.
- **Ceramics and glass -** These items can easily be recycled. Look for special glass recycling bins and bottle banks to dispose them correctly.

3.Organic waste.

- Organic waste is another common household. All food waste, garden waste, manure and rotten meat are classified as organic waste. Over time, organic waste is turned into

manure by microorganisms. However, this does not mean that you can dispose them anywhere.

- Organic waste in landfills causes the production of methane, so it must never be simply discarded with general waste. Instead, look to get a green bin from the Brisbane council, or hire a green skin bin or garden bag for proper waste disposal.

4.Recyclable waste.

- Recyclable rubbish includes all waste items that can be converted into products that can be used again. Solid items such as paper, metals, furniture and organic waste can all be recycled.

- Instead of throwing these items in with regular waste, which then ends up in landfills, place them in your yellow recycling bin or take them to your local Brisbane recycling depot.

- If you're unsure whether an item is recyclable or not, look at the packaging or the diagrams on the lid of your yellow recycling bin. Most products will explicitly state whether they are recyclable or not.

5. Hazardous waste.

- Hazardous waste includes all types of rubbish that are flammable, toxic, corrosive and reactive.

- These items can harm you as well as the environment and must be disposed of correctly. Therefore, I recommend you make use of a waste removal company for proper disposal of all hazardous waste.



Figure 1.1: waste cycling

Chapter 2

Literature review

IOT is playing an important role in growing technologies and there are various studies about many applications.

(1) Kodwo Miezah,Moses Y.

PROJECT TITLE:

Municipal solid waste characterization in Ghana-December 2017

WORK DONE:

Collection of dialy reports on waste collection in Ghana and prepared the reports for the collection efficiency.

CONCLUSION:

Organic waste fraction in the waste was 45 to 69%,National sorting and separation efficiency was 84% for biodegradables and 76% for other waste was separated.

(2)Sehyun park,Sunghoi park

PROJECT TITLE:

IOT based Smart waste management for food-July 2015

WORK DONE:

Collected dialy and weekly reports for food waste and bins are designed to collect large amount of waste.

CONCLUSION:

Web based services are provided for the collection, disposal of food waste. This method improved the collection efficiency of waste collection by 39%.

(3) P.Rajkumar joshi

PROJECT TITLE:

Challenges of municipal waste management-October 2017.

CONCLUSION:

Protection of groundwater contamination from leachate ,developing bio-degradable polythene bags and recycled products.

(4) C.JeyaBharathi

PROJECT TITLE:

Development of an Iot System for Efficient Classification and Management of Solid Waste in Indian Cities-November 2018

WORKDONE:

Developing an effective system for the collection of waste for different zones in city.

CONCLUSION:

This study develops the internet of things practicality based on the management and collection of solid waste for smart city. The automatic sensing system is designed using load cell and ultrasonic sensor to provide an automatic and efficient status of dustbin monitoring system.

(5) Medevdev.J

PROJECT TITLE:

Waste collection system- August 2017

WORKDONE:

Developing the new efficient method for the collection of waste zones with muncipality. CONCLUSION:

Provides greater service quality to smart city citizens. The automatic sensing system is designed using load cell and ultrasonic sensor to provide an automatic and efficient status of dustbin monitoring system.

(6) Navaghane.M,Mahesh.K

PROJECT TITLE:

Waste collection using IR system.- January 2017

WORKDONE:

Using IR sensor to sense the depth and helps the transformation in to various webserver. CONCLUSION:

Monitor the duplicate reports, reduces the corruption in management system, reduces the number of trips of garbage collection vehicle and reduces the overall expenditure related with garbage collection

(7) Vaishali.P,Manoj.T

PROJECT TITLE:

Waste collection using Rasberry Pi.- July 2017

WORKDONE:

Using Rasberry Pi sensor to sense the depth and helps to give information to drivers. CONCLUSION: Maintain the surge of dustbin and make the earth clean and immaculate, reduce the wastage of cost, time and importance of human being and the drivers acquired.

Chapter 3

INTERNET OF THINGS

The Internet of Things (IoT) is the network of physical objects-devices, instruments, vehicles, buildings and other items embedded with electronics, circuits ,software, sensors and network connectivity that enables these objects to collect and exchange data. The Internet of Things allows objects to be sensed and controlled remotely across existing network infrastructure, creating opportunities for more direct integration of the physical world into computer-based systems, and resulting in improved efficiency and accuracy.

- Today, Internet application development demand is very high.
- Basically,IOTis a network in which all physical objects are connected to the internet through network devices or routers and exchange data.
- IOTallows objects to be controlled remotely across existing network infrastructure.
- It refers to the billions of physical devices around the world that are now connected to the internet,all connecting and sharing data.
- The term IOT is mainly used for devices that wouldnt usuallybe generally expected to hacve an internet connection.
- The idea of adding sensors and intelligence to basic objects was discussed in the 1980s and 1990s.
- Processors were cheap and power-frugal enough to be all but disposal where needed became before cost effective.
- The concept of a network of smart devices was discussed as early as 1982, with a modified Coke machine at Carnegie Mellon University becoming the first internet-connected appliance, able to report its inventory and whether newly loaded drinks were cold. Kevin Ashton (born 1968) is

a British technology pioneer who is known for inventing the term "the Internet of Things" to describe a system where the Internet is connected to the physical world viaubiquitous sensors.

- IoT is able to interact without human intervention. Some preliminary IoT applications have been already developed in healthcare, transportation, and automotive industries. IoT technologies are at their infant stages; however, many new developments have occurred in the integration of objects with sensors in the Internet.

- The development of IoT involves many issues such as infrastructure, communications, interfaces, protocols, and standards. The objective of this paper is to give general concept of IoT, the architecture and layers in IoT, some basic terms associated with it and the services provided.

3.1 Concept of IOT

Kevin Ashton firstly proposed the concept of IoT in 1999, and he referred the IoT as uniquely identifiable connected objects withradio-frequencyidentification (RFID) technology. However, the exact definition of IoT is still in the forming process that is subject to the perspectives taken. IoT was generally defined as "dynamic global network infrastructure with self-configuring capabilities based on standards and communication protocols"

- Physical and virtual things in an IoT have their own identities and attributes and are capable of using intelligent interfaces and being integrated as an information network. In easy terms IoT can be treated as a set of connected devices that are uniquely identifiable.

- The words "Internet" and "Things" mean an inter-connected world-wide network based on sensors, communication, networking, and information processing technologies, which might be the new version of information and communications technology (ICT). To date, a number of technologies are involved in IoT, such as wireless sensor networks (WSNs), barcodes, intelligent sensing, RFID, NFCs, low energy wireless communications, cloud computing and so on.

- The IoT describes the next generation of Internet, where the physical things could be accessed and identified through the Internet. Depending on various technologies for the implementation, the definition of the IoT varies. However, the fundamental of IoT implies that objects in an IoT can be identified uniquely in the virtual representations. Within an IoT, all things are able to exchange data and if needed, process data according to predefined schemes.

3.2 Architecture of IOT

A critical requirement of an IoT is that the things in the network must be connected to each other. IoT system architecture must guarantee the operations of IoT, which connects the physical and the virtual worlds. Design of IoT architecture involves many factors such as networking, communication, processes etc. In designing the architecture of IoT, the extensibility, scalability, and operability among devices should be taken into consideration. Due to the fact that things may move and need to interact with others in real-time mode, IoT architecture should be adaptive to make devices interact with other dynamically and support communication amongst them. In addition, IoT should possess the decentralized and heterogeneous nature.

3.2.1 Service oriented architecture

A critical requirement of an IoT is that the things in the network must be inter-connected. IoT system architecture must guarantee the operations of IoT, which bridges the gap between the physical and the virtual worlds. Design of IoT architecture involves many factors such as networking, communication, business models and processes, and security. In designing the architecture of IoT, the extensibility, scalability, and interoperability among heterogeneous devices and their models should be taken into consideration. Due to the fact that things may move physically and need to interact with each other in real-time mode, IoT architecture should be adaptive to make devices interact with other things dynamically and support unambiguous communication of events.

Chapter 4

Methodology&Block Diagram

4.1 Methodology

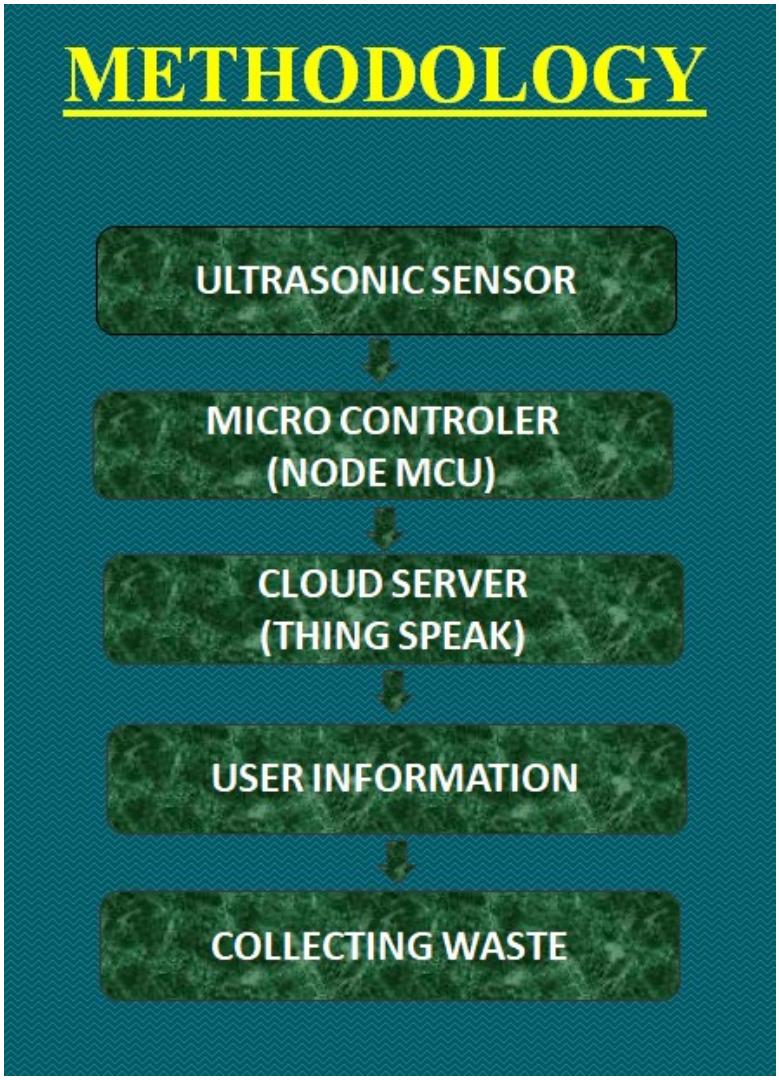


Figure 4.1: Methodology

4.2 Block diagram

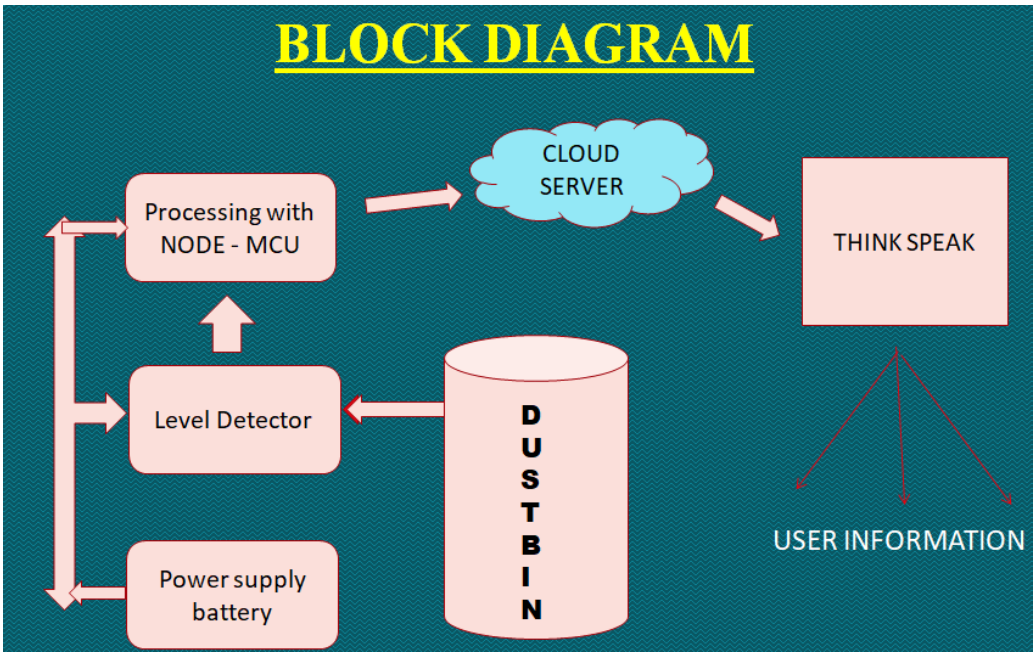


Figure 4.2: Block diagram



Figure 4.3: Block diagram

Chapter 5

Components for waste management

5.1 Components used

1. Ultra sonic sensor.
2. NodeMCU.
3. Batter or power supply.
4. Jamper wires.
5. ThingSPEAK.

1. Ultra sonic senor

- As the name indicates, ultrasonic sensors measure distance by using ultrasonic waves.

- The sensor head emits an ultrasonic wave and receives the wave reflected back from the target. Ultrasonic Sensors measure the distance to the target by measuring the time between the emission and reception.

2. NodeMCU

- Node MCU Development board is featured with WI-FI capability, analog pin, digital pins and serial communication protocols.



Figure 5.1: Ultra sonic sensor

- To get start with using Node MCU for IoT applications first we need to know about how to write/download Node MCU firmware in Node MCU Development Boards. And before that where this Node MCU firmware will get as per our requirement

- It works as a co-ordinator.

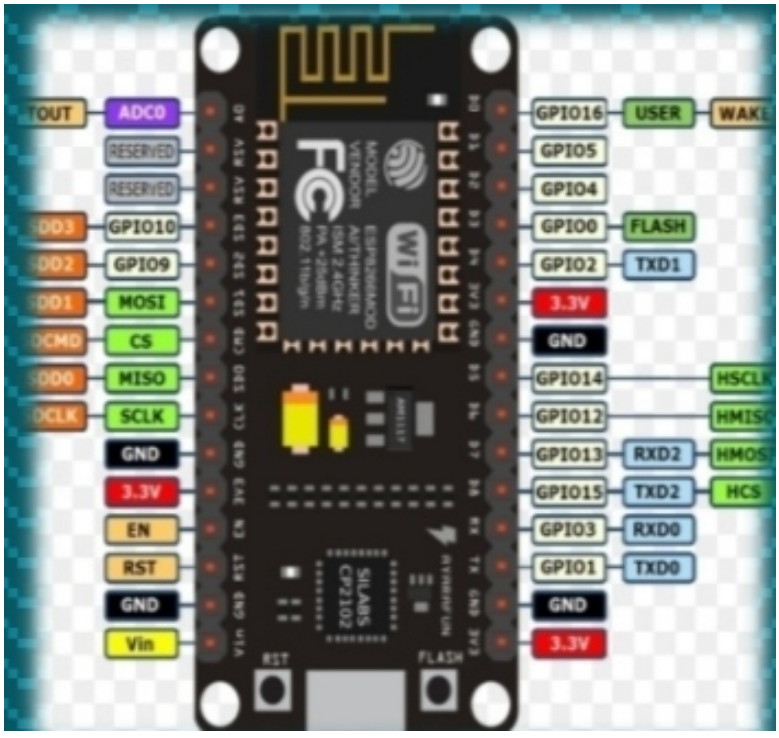


Figure 5.2: NodeMCU

3.Battery

- Power supply to the Arduino

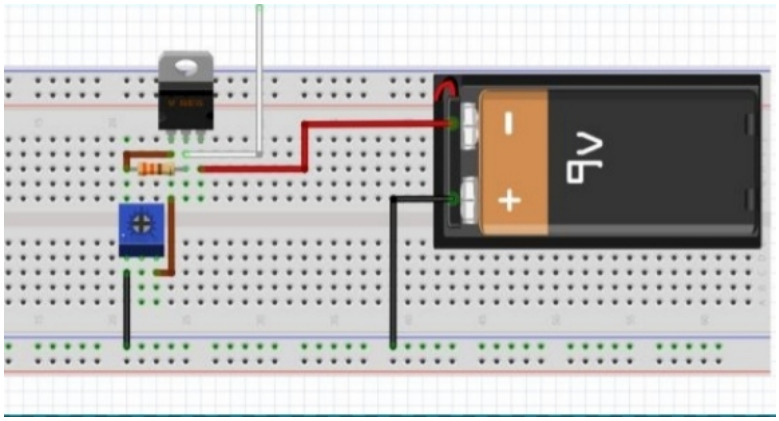


Figure 5.3: Battery

4.Jamper wires

- Used for the connecting NodeMCU,Ultra sonuc sensor and battery

5.ThingSPEAK

- Thing Speakis an Internet of Things (IoT) platform that lets you collect and store sensor data in the cloud and developIoTapplications.

- TheThing SpeakIoTplatform provides apps that let you analyze and visualize your data in MATLAB, and then act on the data.

Chapter 6

Smart waste management using IOT

6.1 Stages in waste management

Our project involves in three stages

1. Sensor work.

2. Fixing the sensor to the garbage bin.

3. Thing speak.

1.Sensor work

- As we discussed above the sensors used are NodeMCU ,ultra sonic sensor.

- Ultra sonic sensor connected to Wi-Fi module i.e., NodeMCU , Ultra sonic sensor sense the depth of the bin and timely inform to the sensor (full/empty).

- As we know that NodeMCU is having Wi-Fi connection it will receives the information and sends the information to cloud server.

2.Fixing the sensors to the garbage bin.

- The sensors are connected at the top of the dust bin , the figures are given below

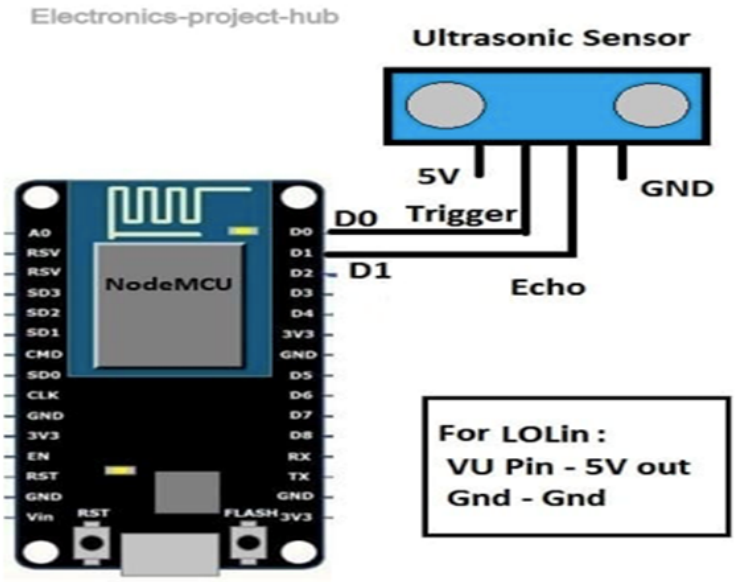


Figure 6.1: Module

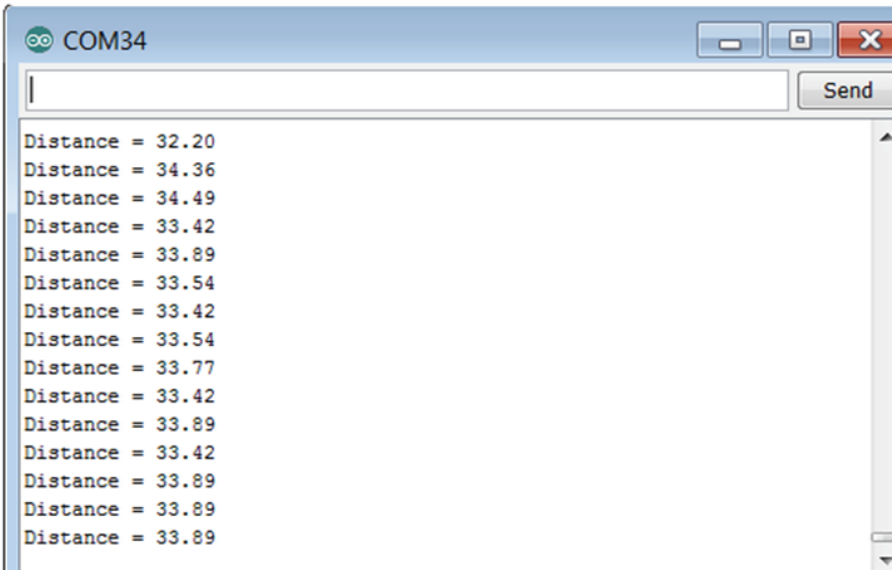


Figure 6.2: Digital values

3.ThingSPEAK.

- All sensor data transferred to the cloud server, it will show the statistics of the bin graphically.

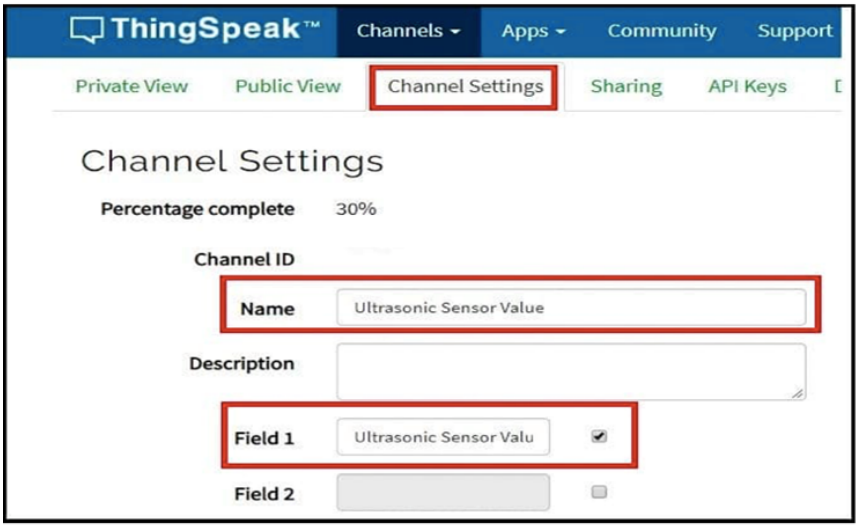


Figure 6.3: Web page

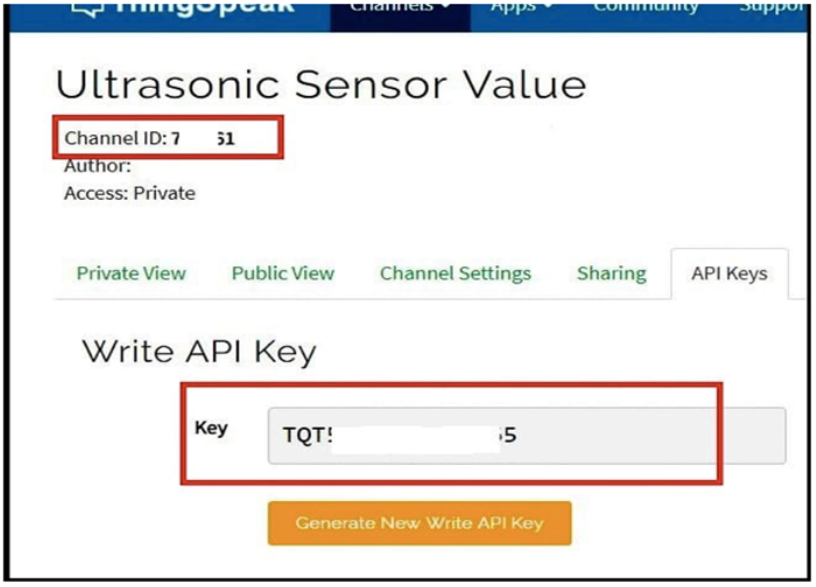
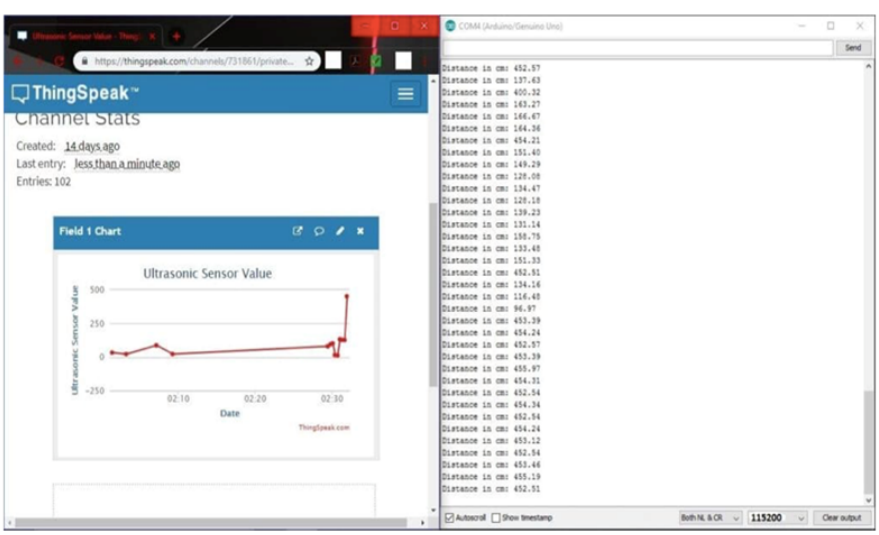


Figure 6.4: Api page



Ultrasonic Sensor data on Thingspeak

Figure 6.5: Statistics of bin

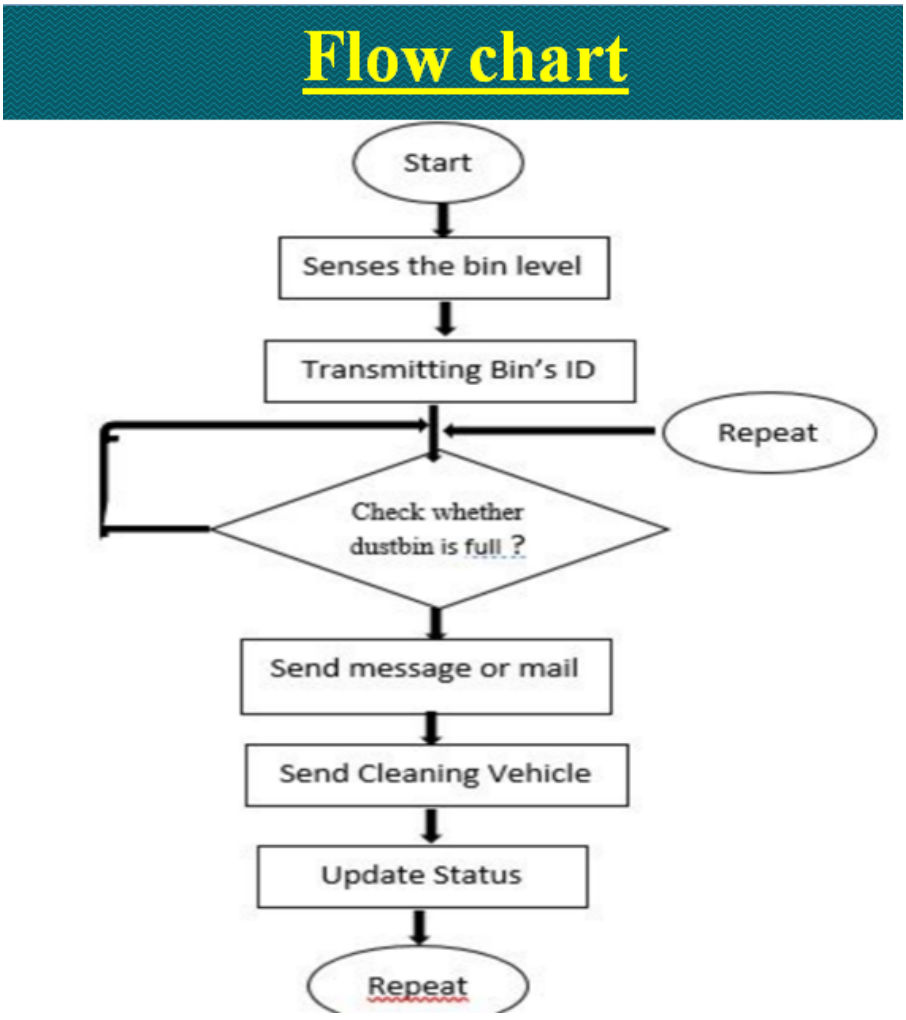


Figure 6.6: Flow chart

```
    # include ”ThingSpeak.h”
# include ¡ESP8266WiFi.h¿
//———— Enter you Wi-Fi Details———//
char ssid[] = ”POCO”; //SSID
char pass[] = ”bharathkumar”; // Password
//————————————————-//
const int trigger = 16;
const int echo = 5;
long T;
float distanceCM;
WiFiClient client;
unsigned long myChannelField = 985327; // Channel ID
const int ChannelField = 1; // Which channel to write data
const char * myWriteAPIKey = ”CU3LH9GYEPAMPKK7”; // Your write API Key
void setup()

Serial.begin(115200);
pinMode(trigger, OUTPUT);
pinMode(echo, INPUT);
WiFi.mode(WIFI STA);
ThingSpeak.begin(client);

void loop()

if (WiFi.status() != WL CONNECTED)

Serial.print(”Attempting to connect to SSID: ”);
Serial.println(ssid);
while (WiFi.status() != WL CONNECTED)

WiFi.begin(ssid, pass);
Serial.print(”.”);
delay(5000);

Serial.println(Connected.”);
```

```
digitalWrite(trigger, LOW);
delay(1);
digitalWrite(trigger, HIGH);
delayMicroseconds(10);
digitalWrite(trigger, LOW);
T = pulseIn(echo, HIGH);
distanceCM = T * 0.034;
distanceCM = distanceCM / 2;
Serial.print("Distance in cm: ");
Serial.println(distanceCM);
ThingSpeak.writeField(myChannelField, ChannelField, distanceCM, myWriteAPIKey);
delay(1000);
```

Chapter 7

Advantages & Disadvantages

7.1 Advantages

- A reduction in the number ofwastecollections needed by up to 80%, resulting in less manpower, emissions, fuel use and traffic congestion.

- Cost reduction , Our smart waste logistics solution reduces waste collection frequency dramatically, which enables you to save on fuel, labor, and fleet maintenance costs.

- Improved cleanliness , In densely populated areas, a rapid waste generation often leads to overflowing waste bins and unsightly streets. Our solution enables waste collection staff to read fill-levels in real time and receive notifications of waste overflows.

- CO2 reduction , Collecting garbage is a very pollutant heavy proposition. Our solution offers you the means to have less trucks on the road for less time, which means less greenhouse gas emissions, less noise pollution, and less road wear.

7.2 Disadvantages

- System requires more number ofwastebins for separatewaste.
  collectionas per population in the city. This results into high initial cost due to expensivesmart-dustbins compare to other methods.

- Sensor nodes used in the dustbins have limited memory size.

- The trainining has to be provided to the people involved in the smart waste management system.

Chapter 8

Conclusion

- The structure depends upon IoT recognizing model.

- The status of the bin has recorded graphically in cloud server

- The information of the bin(full or empty) given to the drivers from the main office in the town.

- The execution of waste association framework by utilizing sharp dustbins to check the segment of impressive dustbins paying little notice to whether the dustbin are full or not.

- In this framework when garbage is full the data is send to the insisted individual.

- By executing this proposed framework we can build up the sharp city thought and cost is reduced.

- By the productive utilization of sharp dustbins can the advantage is advanced. This framework diminishes the improvement in the awe inspiring city, with the target that condition will be cleaned.

Chapter 9

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